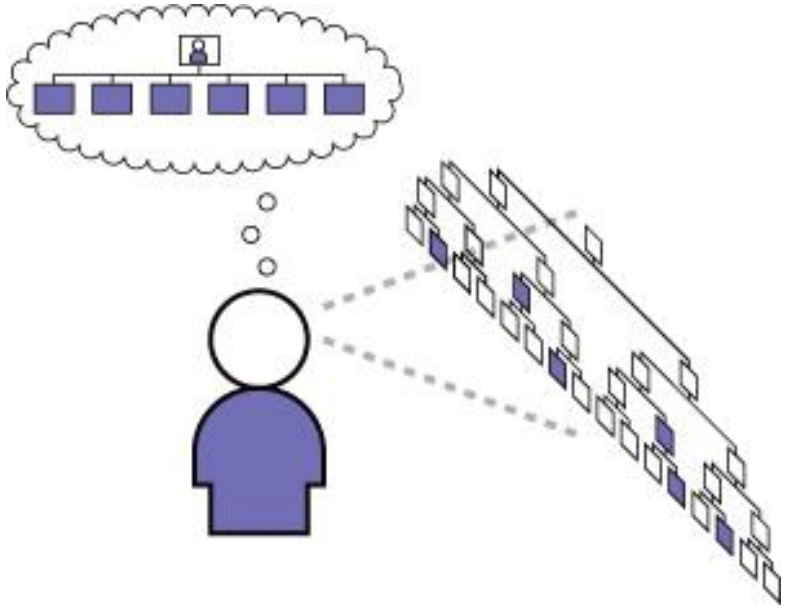
User Interfaces

Designing Effective Human-System Interaction

FINTECH 512 – SOFTWARE ENGINEERING

What Is User Experience?

“The experience a product creates for the people who use it in the real world.”
- Jesse James Garrett



'The Elements of User Experience: User-Centered Design for the Web and Beyond’ – Jesse James Garrett

User Experience (UX)

An aspect of the 'usability' quality attribute

”User interface design should ensure that interaction between the human and the machine provides for effective operation” [0]

“If a software system is to achieve its full potential, it is essential that its user interface should be designed to match the skills, experience and expectations of its anticipated users.” [1]

“Usability principles can provide guidelines for discovering problems in the design of the user interface” [0]

[0] SWEBOK
[1] https://ifs.host.cs.st-andrews.ac.uk/Books/SE9/WebChapters/PDF/Ch_29%20Interaction_design.pdf

UX Elements: Five Layers('Planes')

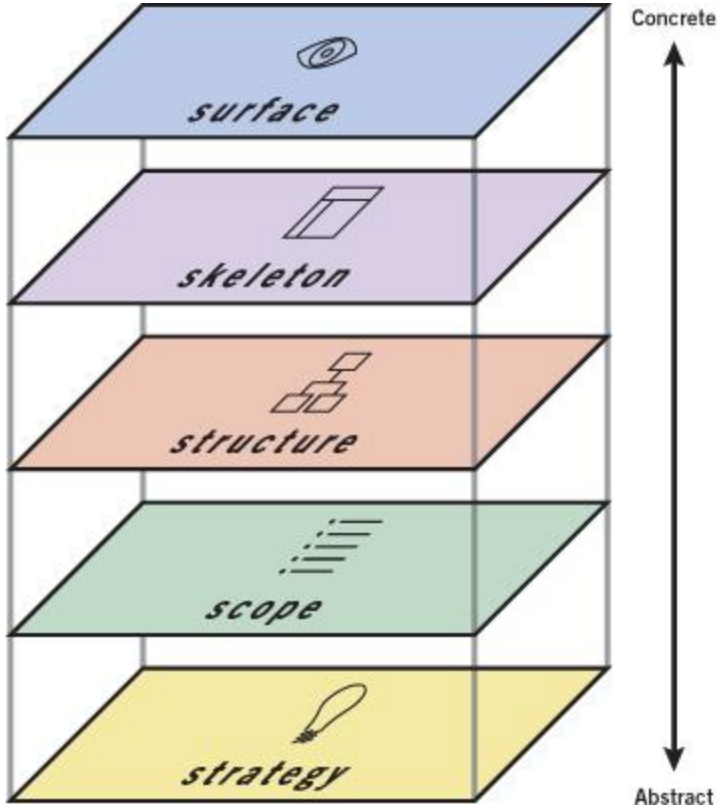
Surface – images, text, input controls

Skeleton – placement of images, text, controls (e.g. on a page)

Structure – navigation (e.g. between pages)

Scope – features and functions of the product

Strategy – what the users and operators of the product hope to accomplish



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Today: User Interface (UI)

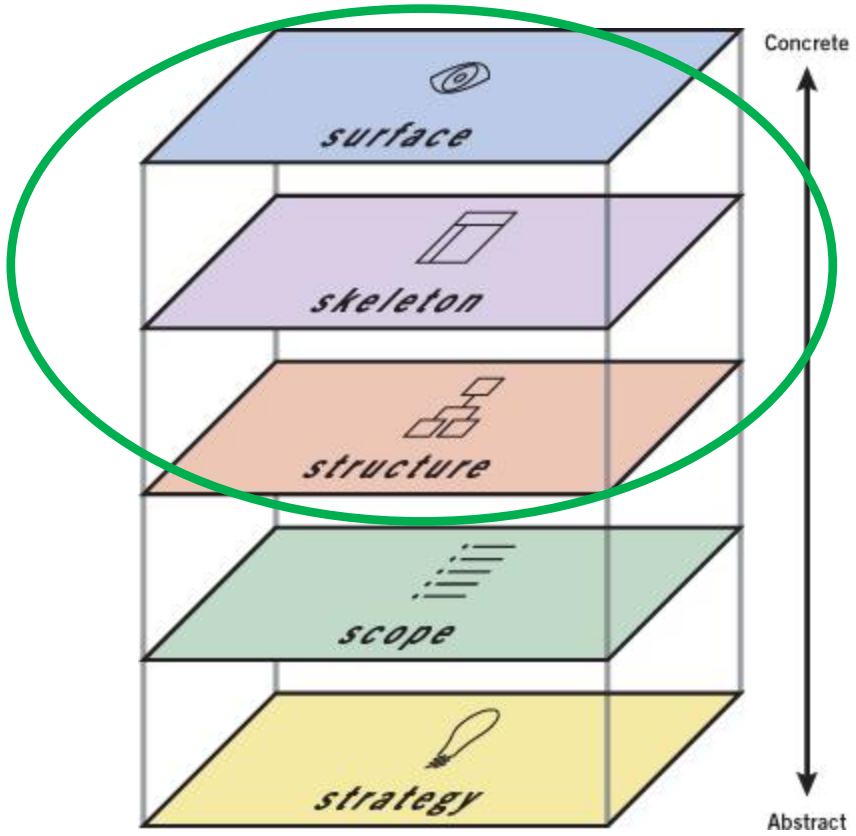
Surface – images, text, input controls
(UI Design, HTML, CSS, JS)

Skeleton – placement of images, text, controls
(Use Cases, UI Design, Flask templates, HTML, CSS, JS)

Structure – organization, navigation
(Use Cases, Design, Patterns, Flask Routes)

Scope – features and functions of the product
(Use Cases, Design)

Strategy – what the users and operators of the product hope to accomplish
(Requirements, User Stories)



'The Elements of User Experience: User-Centered Design for the Web and Beyond' – Jesse James Garrett

Stock Symbol: Submit

Company Name: Apple Inc

Stock Symbol: AAPL

Exchange: NASDAQ

Sector: TECHNOLOGY

Industry: ELECTRONIC COMPUTERS

Market Cap: $2815367315000

PE Ratio: 28.35

Earnings Per Share: $6.43

Dividend per Share: 0.95

Dividend Yield: 0.0053

52 Week High: $199.37

52 Week Low: $143.14

Stock Data

Current Price: $182.3200

Previous Close: $181.5600

Open: $181.9400

Volume: 41529674

Recent News Articles

11 common financial mistakes - and how to avoid them

Learn from these pitfalls ...

Apple Says It Has Rebuilt The iMessage Cryptographic Protocol From 'Ground Up' To Thwart Quantum Computing Threat - Apple (NASDAQ:AAPL)

In a bid to fortify its messaging platform against potential quantum computing decryption, Apple Inc. AAPL has introduced a new security upgrade for iMessage. What Hap
The new security protocol, dubbed PQ3, is a significant overhaul of the iMessage cryptographic protocol.

Why Apple's new Sports app could be bad news for Google

One expert said Apple's Sports app might eventually be part of the built-in apps for iPhones, like Maps and Notes ...

Apple Dominates 2023 Smartphone List With 7 Of Top 10 Models - Samsung Electronics Co (OTC:SSNLF) , Apple (NASDAQ:AAPL)

Apple Inc AAPL has seven of the top 10 selling smartphone models by units shipped. Here were the top 10 smartphone models by market share in 2023, according to Counterpoint Research. The findings are in line with a previous report from analytics firm Canalys.

Stock Ticker AAPL Search

Apple Inc (NASDAQ:AAPL)

Industry: Electronic Computers
Sector: Technology

Current Price: $171.21	Opening Price: $172.02	Previous Closing Price: $170.69
Volume Traded: 51861083 units	52-week High: $199.37	52-week Low: $143.14

Market Cap:	**$2.82T**
P/E Ratio:	**28.35%**
EPS:	**6.43%**
High:	**$173.07**
Low:	**$170.34**
Dividend & Yield:	**0.01%**

Latest News

11 common financial mistakes - and how to avoid them
Learn from these pitfalls ...

Jim Cramer Evaluates Nvidia, Amazon And Other Magnificent 7 Earnings: 'The Seven Truly Did Acquit Themselves Well' - Apple (NASDAQ:AAPL

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What Different Kinds of UIs Exist?

Class discussion:

Describe three or more ways to buy a beverage

What Different Kinds of UIs Exist?

Class discussion, part 2:

For each approach:

- What works well for that user interface?
- What does not work well?
- Who are the target users?
- What technology is required for that user interface?
- In what environment does that user interface work best?

What Different Kinds of UIs Exists?

Graphical

Text / Terminal

Mobile

Web

Command (aka “voice”)

Command Line

Phone

What Makes a Good User Interface?

https://www.interaction-design.org

What Different Kinds of UIs Exists?

Graphical

Text / Terminal

Mobile

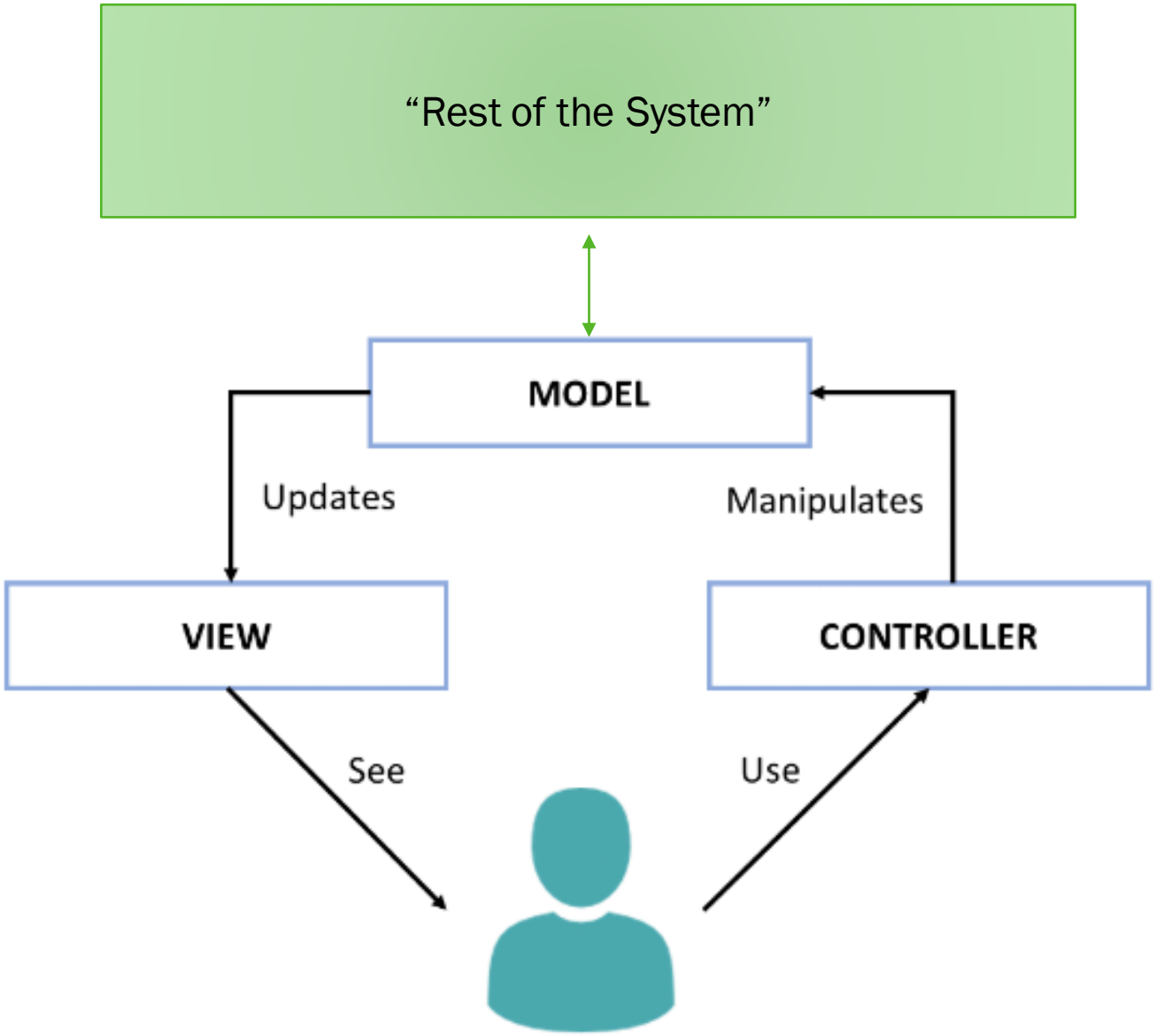
Web

Command (aka “Voice”)

Command Line

Phone

From a software design / implementation perspective, what is common across these different types?



MVC Design Pattern

MVC Pattern

Software architecture design pattern

Supports parallel development

Loose coupling between model and view increases flexibility in use

- e.g., multiple views for one model

Separates functionality (Single Responsibility)

Web Applications:

Site Structure

Navigation

Content

Let’s look at the following sites:

- How do they implement each of these?
- What are they prioritizing?

User ID

Password

Save User ID

Log In

Forgot ID/Password?

Security & Help | Enroll

Open an Account

Find your closest financial center or ATM

Schedule an Appointment

Choose the card that works for you

Customized Cash Rewards	Unlimited Cash Rewards	Travel Rewards	BankAmericard®
No annual fee.	No annual fee.	No annual fee.	No annual fee.
$200 online bonus offer	$200 online bonus offer	25,000 online bonus points offer	0% intro APR offer
3% cash back in the category of your choice >	Unlimited 1.5% cash back on all purchases >	Unlimited 1.5 points for every $1 spent on all purchases >	Intro APR offer for 18 billing cycles >

Open a checking account

Consider the benefits of opening a Bank of America Advantage Banking Account.

What if you really listened to the kid in you?

Jeni, of Jeni's Splendid Ice Creams, on what

Information on assistance programs and the latest status on our locations and safety measures. Learn more >

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Personal | Investing & Wealth Management | Small Business | Commercial Banking | Corporate & Investment Banking

Banking and Credit Cards | Loans and Credit | Investing and Retirement | Wealth Management | Rewards and Benefits

View Your Accounts

Username

Password

Save username

Sign On

Forgot Password/Username?

Enroll Now
Security Center
Privacy, Cookies, and Security

$0 has never felt like so much

$0 online trades[1] with *WellsTrade®*

Take Control >

Simplify life: Everyday Checking | Find your credit card | Free online budgeting tools | Make an appointment | Check today's rates

See how we can help you achieve your goals

Explore your credit options

Improve your credit >

Borrowing and Credit

Duke TODAY

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Duke Creates Krzyzewski Family Scholarship to Honor Legendary Coach

Baby Receives World's First Combination Heart Transplant-

ATHLETICS

50 Years of Title IX

Since 1972, the Wolfpack has paved the way for equality and access.

Replay the history →

THE UNIVERSITY
of NORTH CAROLINA
at CHAPEL HILL

DISCOVER NAVIGATE SEARCH

The next chapter

Nearly 200 Tar Heels at the UNC School Medicine will take a giant step forward in their careers this week as learn where they will work as resident physicians for the next several years.

LEARN MORE ABOUT MATCH DAY

VISIT SUPPORT APPLY COVID-19 RESOURCES

User Experience (UX)

Just because we can make a GUI does not mean that it is **good**

- We want people to be able to use it well: have a good User Experience

How many of you have ever used software with a badly designed user interface?

- What did you think?
- Did you keep using the software?

Is this GUI good?

Why or why not?

Does it matter?

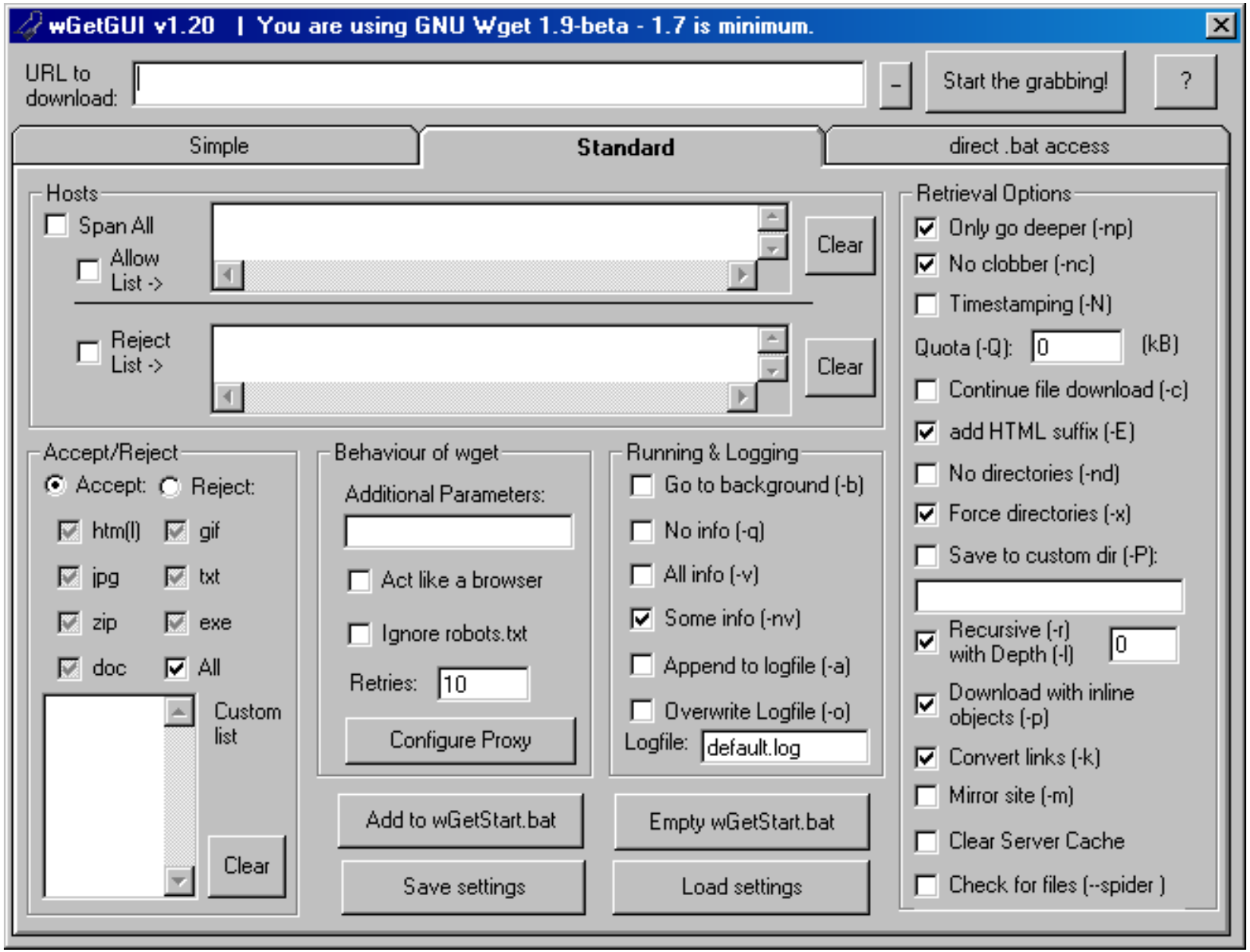


FIGURE 1

	BDLL	Borrower LIBOR Drawdown Prod	Drawdown
	001BDLL201480094		001BDLL201480094
	024462	REVLON CONSUMER PRODUCTS CORP	
Facility Name	REVLON TERM LOAN 2016		

GL Detail

Component	Internal GL	Overwrite default settlement instruction
COLLAT		☐
COMPINTSF		☐
DEFAUL		☐
DFLFTC		☐
FRONT		☐
FUND		☐
INTEREST		☐
PRINCIPAL	3003000023	☑
		☐

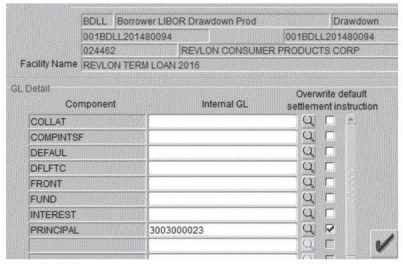
https://www.bloomberg.com/opinion/articles/2021-02-17/citi-can-t-have-its-900-million-back

“Citigroup trial reveals chain of gaffes that led to the $900 million blunder”

“Referred to as one of the biggest blunders in banking history, the mistake hinges on Citibank’s failure to properly execute its “six eyes protocol” that requires three people to review and execute wire transfers from its transitional finance group.”

“The bank’s loan processing program, Flexcube, has a default of the wire payment unless one of the members of the six eyes protocol overrides the option, an action that did not occur in this case. Known as a “fat-finger mistake,” the term refers to a human error made by pressing the wrong key”

FIGURE 1



https://prmia.org/common/Uploaded%20files/ORM%20Designation/PRMIA_Citibank-Revlon_021122.pdf

Bad UX Kills Products

Estimated **17% of startups fail** because of user experience issues

"Bad looking or poorly designed websites, **create mistrust** among users leading to higher drop-offs and almost no returning customers. 75% of user trust is gained just by a well-designed website."

"**39% of people will stop engaging** with a website if the images take a long time to load. The attention span of humans is falling short every day and if you're building heavy sites with slow servers, you are going to be losing a lot of customers."

"**95% of mobile applications are uninstalled within a month**. The users find that once the instant gratification is met, there is no further experience that motivates the user to stick around."

"**88%** of the online customers are **less likely to return** to a website or an app once they have encountered a bad experience."

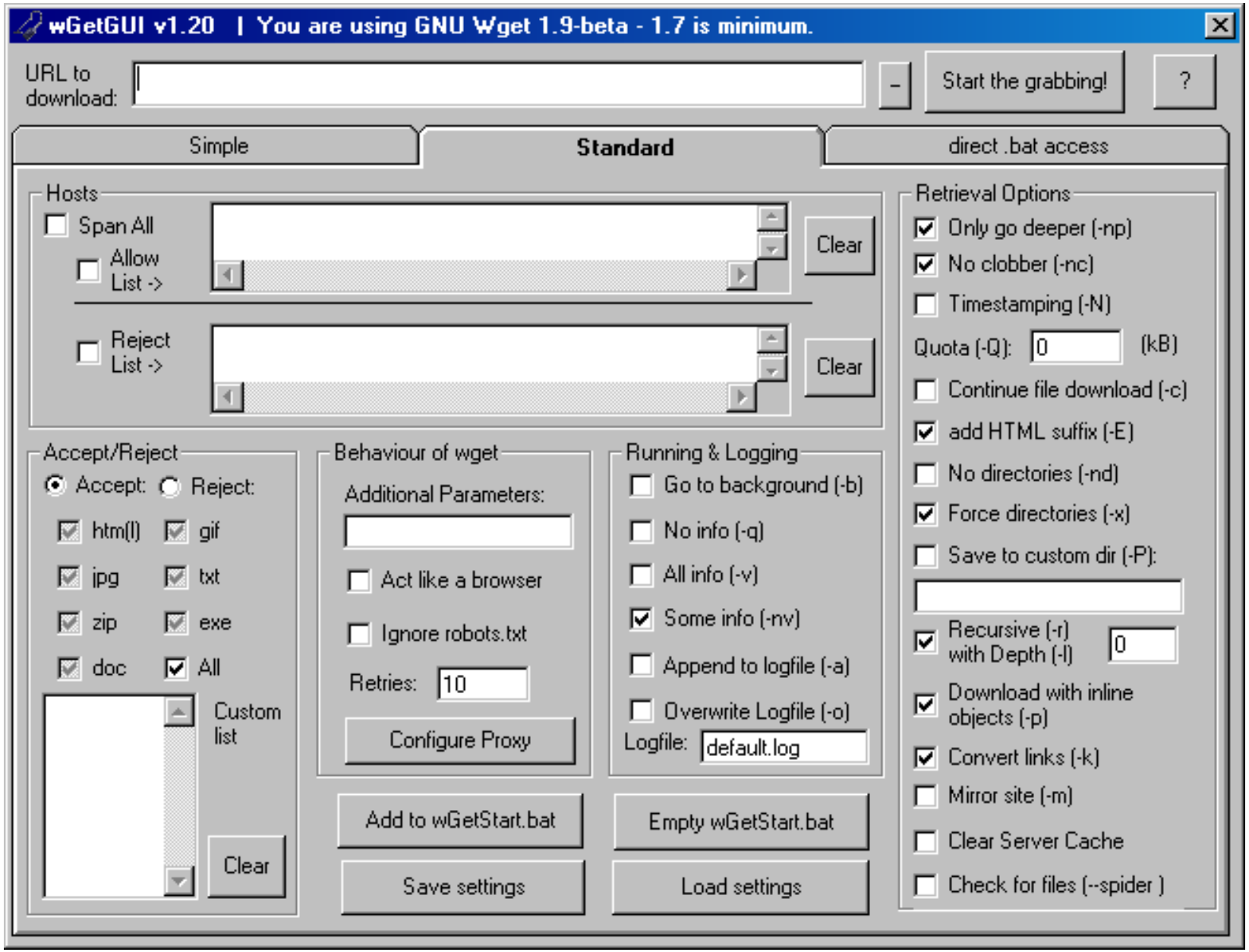
Source: https://www.impactbnd.com/blog/user-experience-stats-infographic

“No product is an island. A product is more than the product. It is a cohesive, integrated set of experiences. Think through all of the stages of a product or service – from initial intentions through final reflections, from first usage to help, service, and maintenance. Make them all work together seamlessly.”

— DON NORMAN, INVENTOR OF THE TERM “USER EXPERIENCE”

One problem here:
High cognitive load

Cognitive load: how much working memory your brain needs to use



Some UX terminology

Cognitive load: how much working memory your brain needs to use

- 7 item limit

Affordance: what it looks like you can do with something

- Analogous to least surprise?

Hidden State: invisible information in the system that affects your interaction.

- Vim: does :wq save and quit? Or enter the text ":wq" in your document? Depends on the mode…
- Powerpoint tells me it can't save due to some font. No idea where it is…

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The Magical Number Seven, Plus or Minus Two
Some Limits on Our Capacity for Processing Information

George A. Miller
Harvard University

This paper was first read as an Invited Address before the Eastern Psychological Association in Philadelphia on April 15, 1955. Preparation of the paper was supported by the Harvard Psycho-Acoustic Laboratory under Contract N5ori-76 between Harvard University and the Office of Naval Research, U.S. Navy (Project NR 142-201, Report PNR-174). Reproduction for any purpose of the U.S. Government is permitted.
Received: May 4, 1955

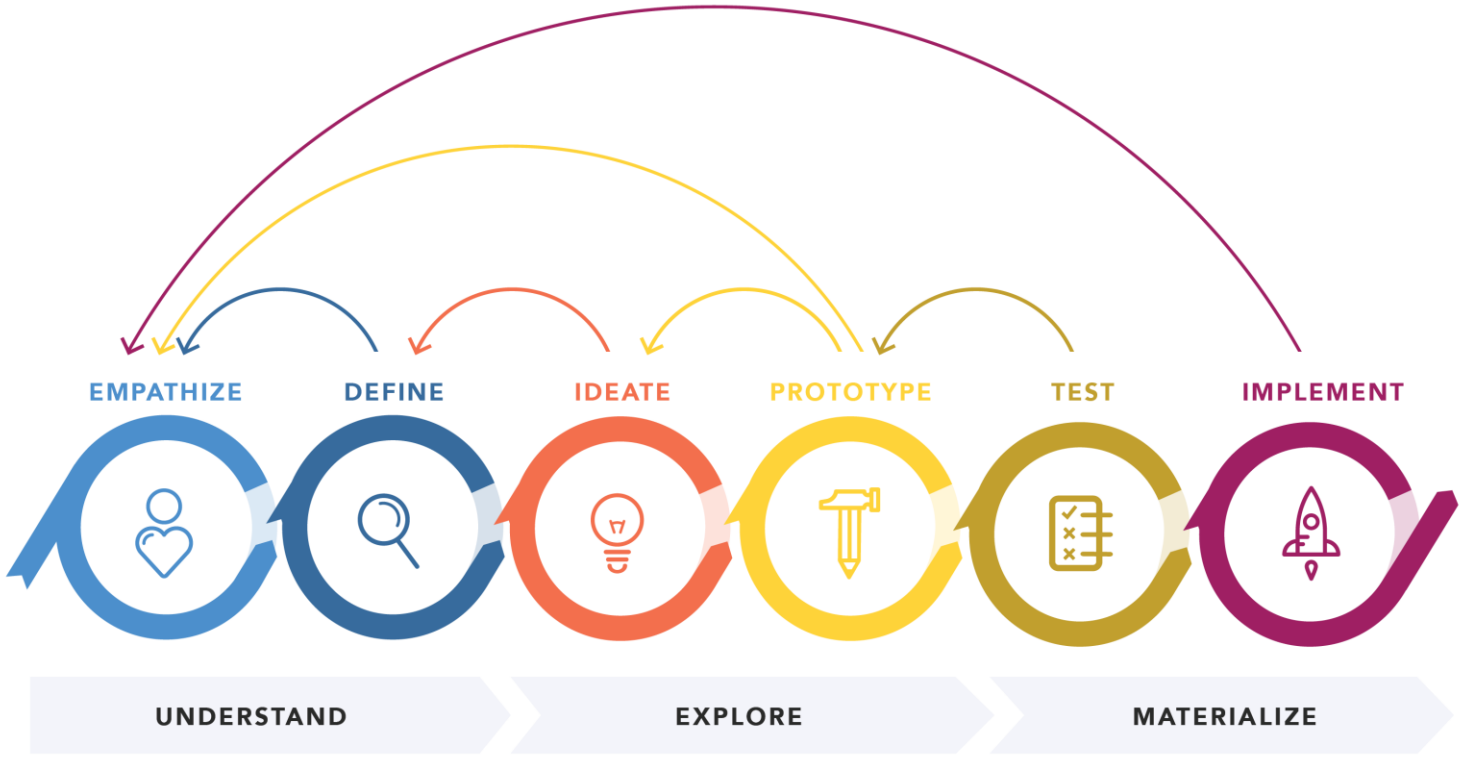
My problem is that I have been persecuted by an integer. For seven years this number has followed me around, has intruded in my most private data, and has assaulted me from the pages of our most public journals. This number assumes a variety of disguises, being sometimes a little larger and sometimes a little smaller than usual, but never changing so much as to be unrecognizable. The persistence with which this number plagues me is far more than a random accident. There is, to quote a famous senator, a design behind it, some pattern governing its appearances. Either there really is something unusual about the number or else I am suffering from delusions of persecution.

http://www2.psych.utoronto.ca/users/peterson/psy430s2001/Miller%20GA%20Magical%20Seven%20Psych%20Review%201955.pdf

UX Design

“the process design teams use to create products that provide meaningful and relevant experiences to users” – Interaction Design Foundation

UX Methodology



Process Overview

- Collect User data
- Determine customer pain points / goals
- Personas
- Ideation (create solutions)
- Scenarios & Storyboards
- “Paper” prototypes
- Usability testing

Prototyping(1)

What?

- Early working model of a design used to gain **rapid** feedback

Why?

- Communicate with team (Proactive)
- Limit expense of development (reducing risk of technical debt)
- Usability testing - test flows and interactions (e.g, buttons make sense, actions are what expect)
- Validate new ideas (determine if worth investment prior to resources)
- Rapidly iterate through ideas

Types

- Sketches, Wireframes, Mockups
- Usually do NOT use code

Important: you want
to know if someone other than you can use this!

Prototyping(2)

Fidelity: visual appearance and the degree of functionality represented in a prototype

- Low, Medium, High
- Make sure fidelity matches what feedback you're interested in

Sketch (low)

- Drawing of the user interface

Wireframes (low-medium)

- Representation of information and overall layout; grayscale/black and white

Mockups (medium –high)

- Visual design including color, hierarchy, and iconography

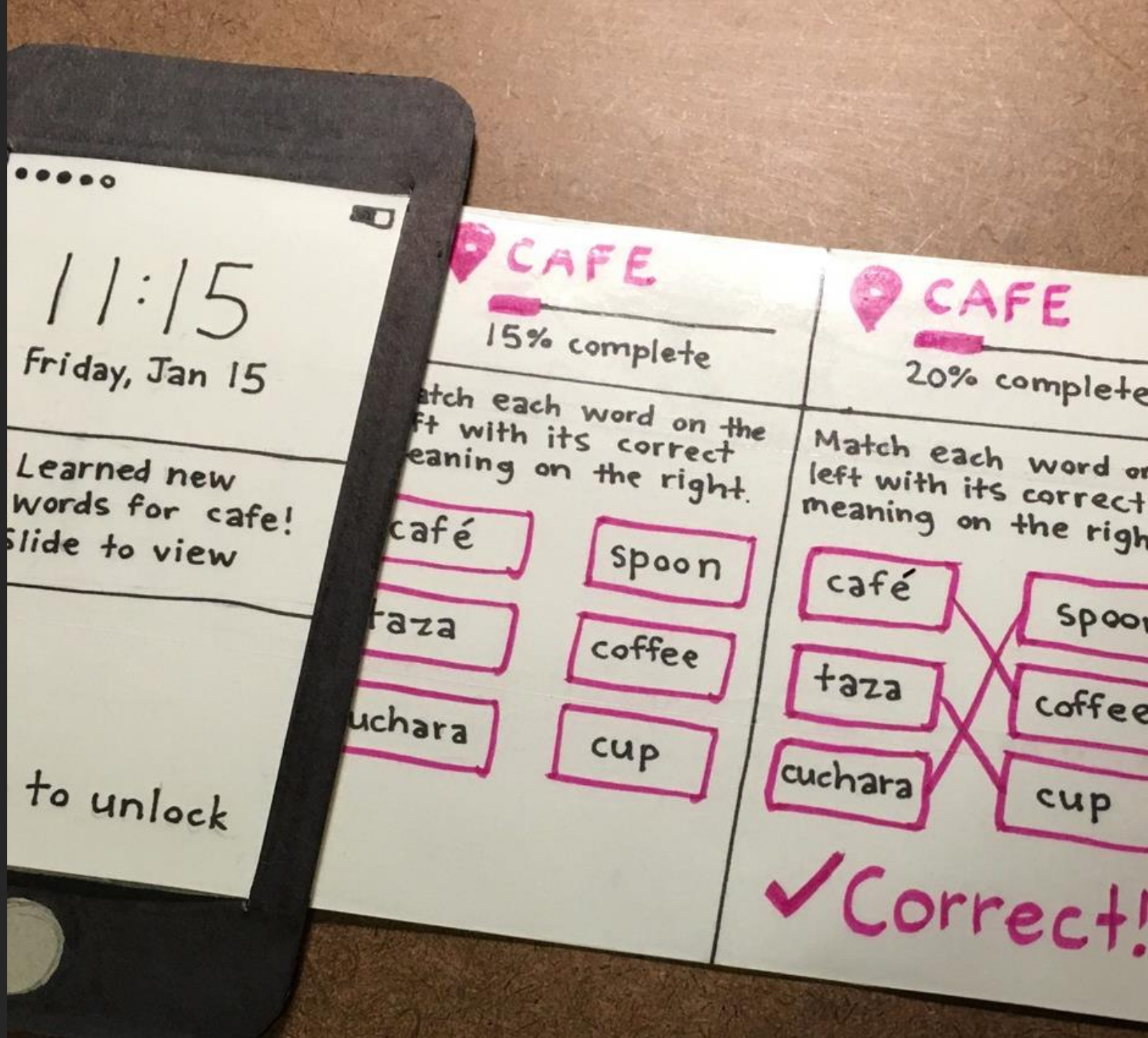
Paper Prototypes

Low fidelity

Focused on layouts, lightweight interactions, and flows

Use: markers, paper, construction, sticky notes

Requires facilitator to support exploration



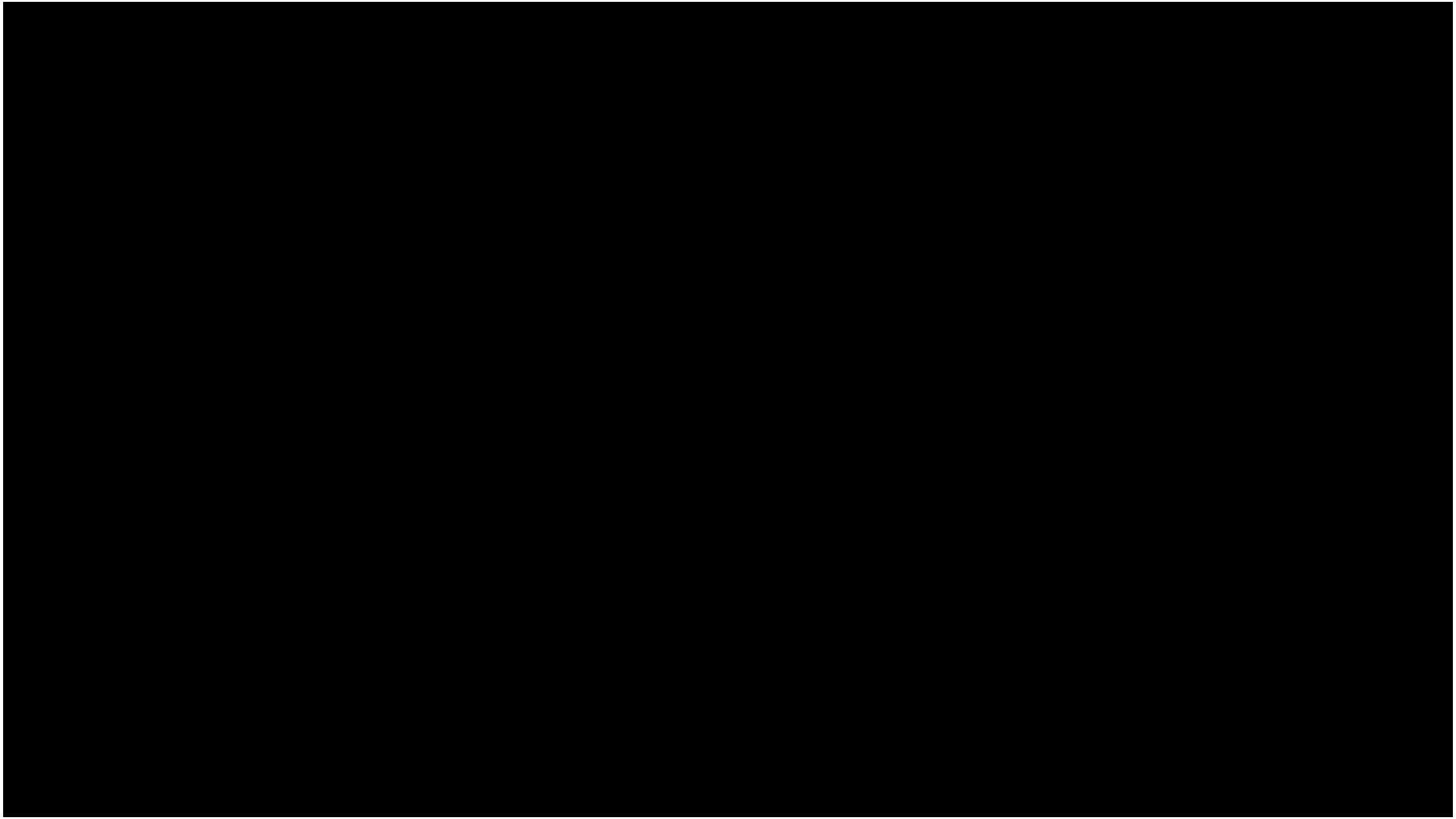
Think, Pair, Share

What do you notice?

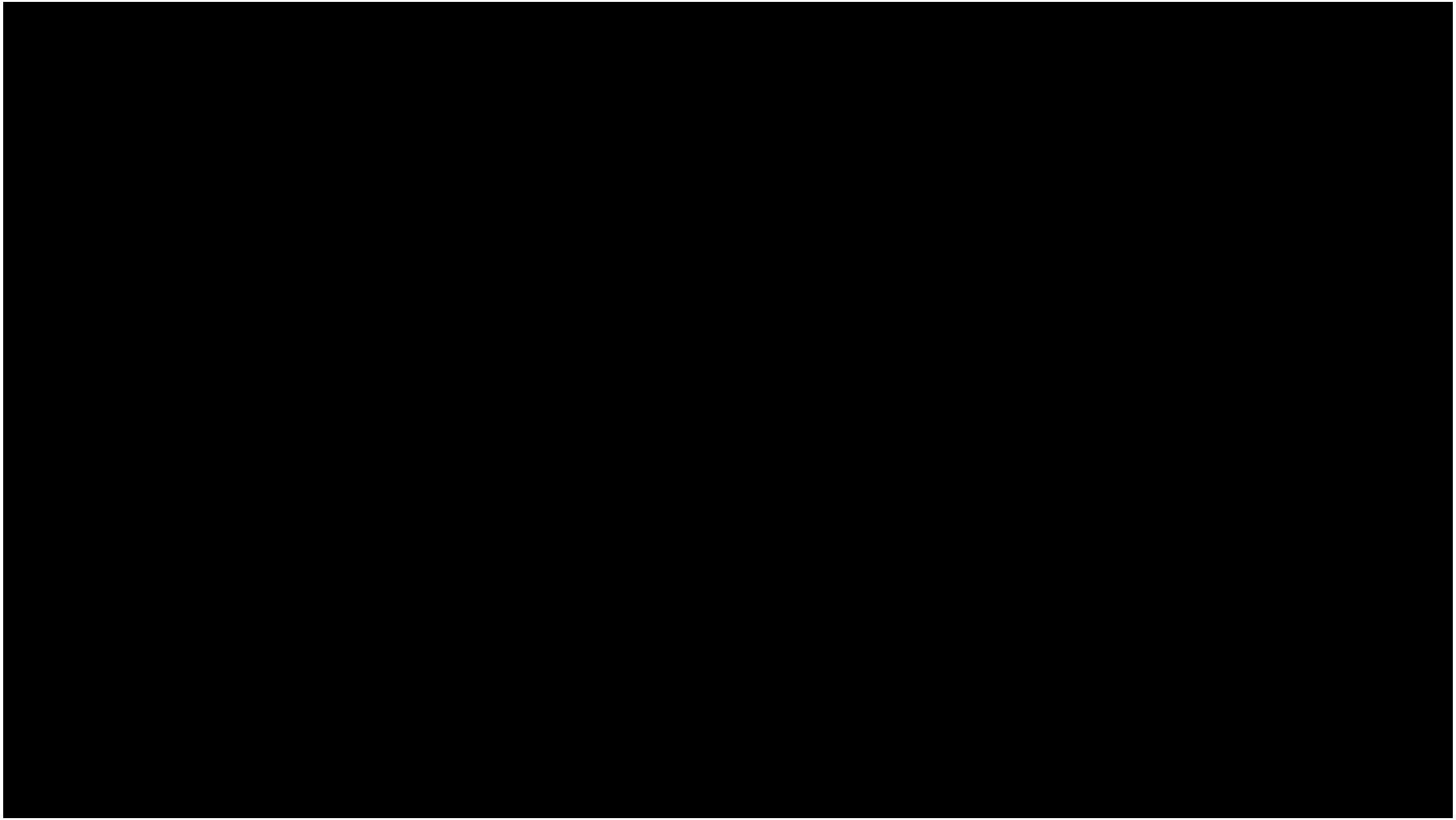
What are differences between the approaches?

What did you like?

What seems confusing?



Mobile Application Design: Paper Prototype Video - https://www.youtube.com/watch?v=y20E3qBmHpg



Low Fidelity Prototype Testing of the EE App: https://www.youtube.com/watch?v=yafaGNFu8Eg

Think, Pair, Share

What do you notice?

What are differences between the approaches?

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What seems confusing?

User Testing with Paper

At least 2 people

- Facilitator – communicates tasks to complete
- “Computer” – changes prototype in response to user actions

Tasks

- Things you’d expect the real system to do
- Limited by what prototype can actually do

Paper Prototype Benefits

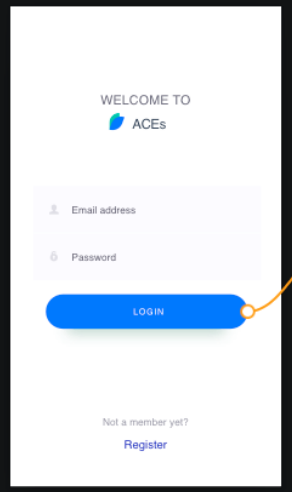
Very early feedback on concepts by users

Users typically more open to provide pushback on design (paper not ”fancy”)

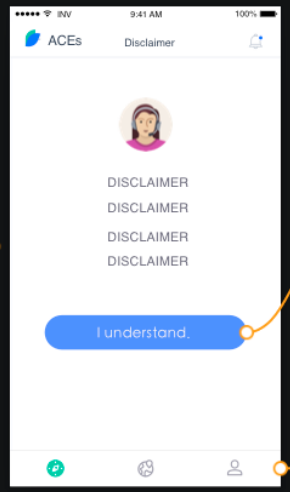
Testing design of **interaction** not the **visual** design

“Paper” Protoyping Tools

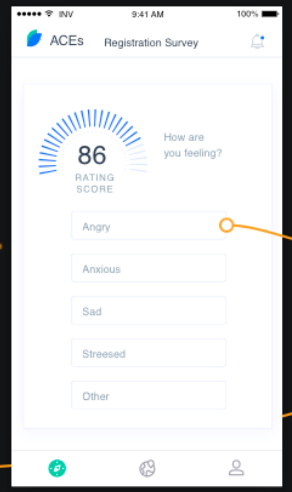
ACEs - Login



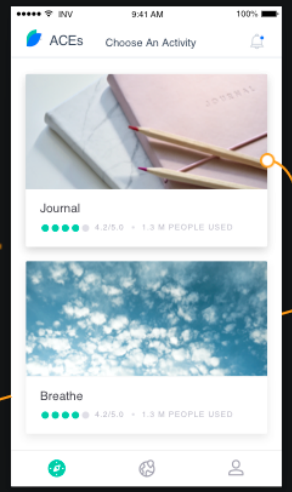
ACEs - disclaimer



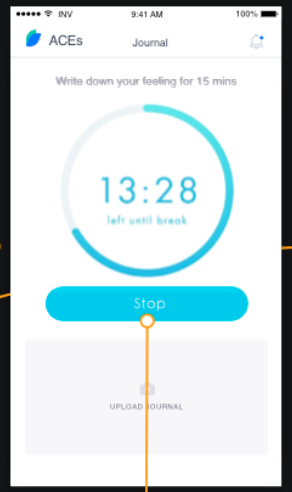
ACEs - registration



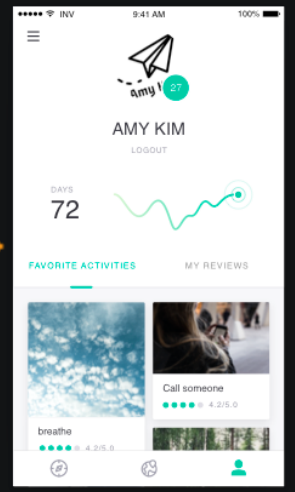
ACEs - activities



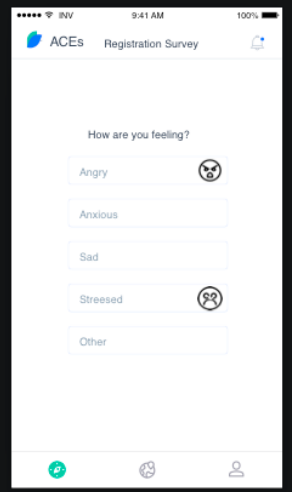
ACEs - Journal



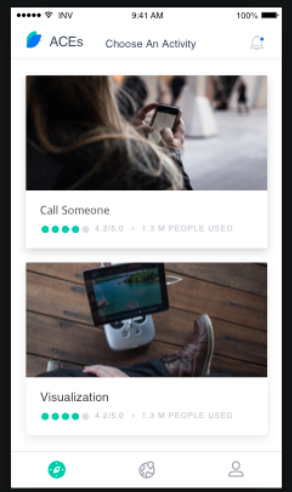
USER PROFILE



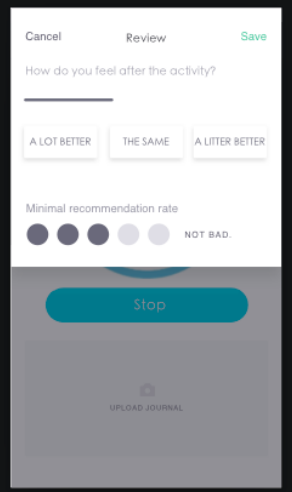
ACEs - re...tration (1)



ACEs - activities (1)



ACEs - feedback



Build With Standard Components

As you put your prototype together, use standard components for standard purposes:

Click on something? Button
Adjust something? Slider
Yes/No? Checkbox
Select One of many: Radio Buttons or Drop Down List
....

See https://www.usability.gov/how-to-and-tools/methods/user-interface-elements.html

Also: think about affordance!

Bootstrap: https://getbootstrap.com/

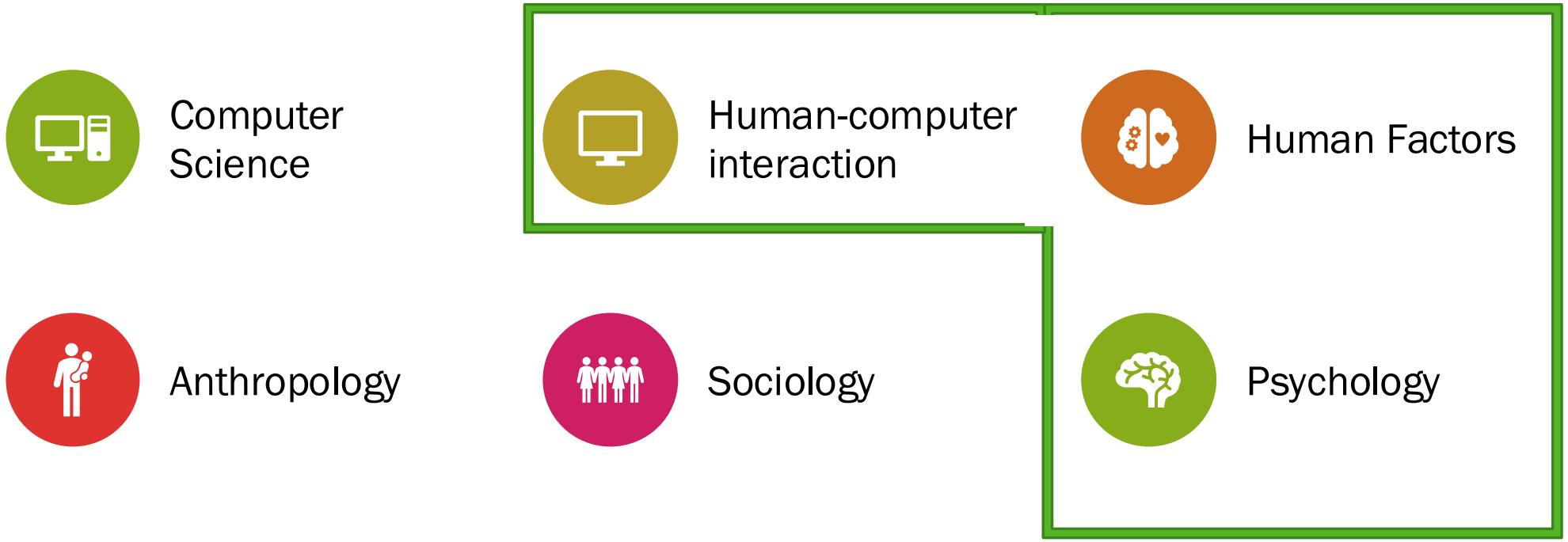
Usability

- **Learnability**: How easy is it for users to accomplish basic tasks the first time they encounter the design?
- **Efficiency**: Once users have learned the design, how quickly can they perform tasks?
- **Memorability**: When users return to the design after a period of not using it, how easily can they reestablish proficiency?
- **Errors**: How many errors do users make, how severe are these errors, and how easily can they recover?
- **Satisfaction**: How pleasant is it to use the design?

https://www.nngroup.com/articles/usability-101-introduction-to-usability/

Measuring usability…

1. Success rate (whether users can perform the task at all)
2. Time a task requires
3. Error rate
4. Users' subjective satisfaction

https://www.nngroup.com/articles/usability-metrics/

Many Relevant Fields